

GEORGIA INSTITUTE OF TECHNOLOGY
COE3001-H MECHANICS OF DEFORMABLE BODIES

Midterm exam - 2026 Spring

Mar. 4, 8:00-9:15 am

Name: _____ GTID: _____

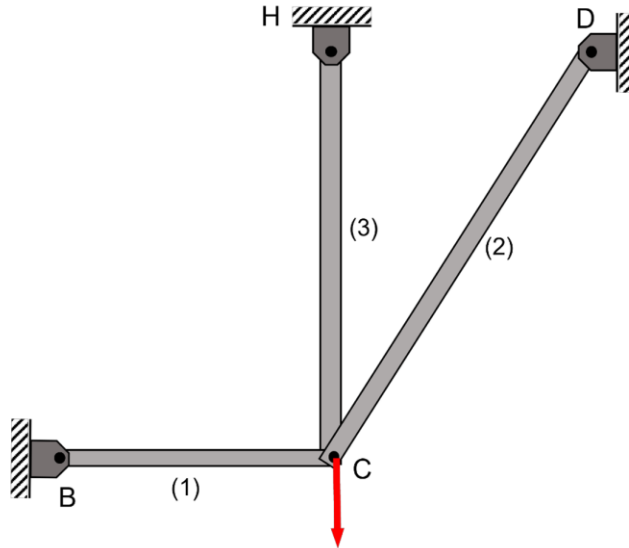
Please read the following instructions

- ✧ You have **75** minutes to complete this exam.
- ✧ You may use a single, double-sided, letter sized formula sheet.
- ✧ This question booklet contains **6** questions, **11** pages (including the cover) for a total of **30** points. Check to see if any pages are missing.
- ✧ DO NOT scribble or do rough work or make any stray marks on it. Use separate sheet for rough work.
- ✧ All the questions are compulsory, and all the notations have their usual meaning.
- ✧ Show all necessary steps for full credit. Clearly indicate your final answers.

Problem 1 (4 points):

Consider the truss shown in the figure. Assume small deformations and linear elastic behavior for all members. Circle the correct answer.

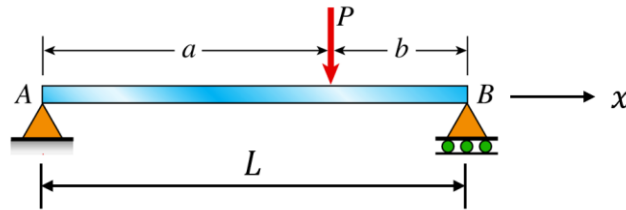
- (1) The structure is *statically determinate* or *statically indeterminate*. (2 pts)
(2) *True* or *False*: The stress in element (1) depends on the material of (2) & (3). (2 pts)



(Partial credit will be given for showing your work)

Problem 2 (4 pts)

Consider the simply supported beam shown in the figure. A concentrated load P is applied at a distance a from support A and b from support B . The total length of the beam is $L = a + b$.



- (1) What is the magnitude of the internal shear force in the segment between support A and the applied load P in terms of a , b , and P ? _____ (1 pt)
- (2) What is the magnitude of the internal shear force in the segment between the applied load P and support B in terms of a , b , and P ? _____ (1 pt)
- (2) Which section of the beam experiences the maximum bending moment magnitude?
 - (A) At support A ,
 - (B) At support B
 - (C) Directly under the applied load P
 - (D) At midspan $x = L/2$

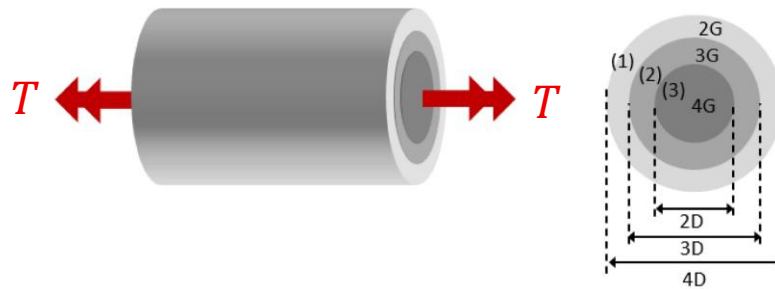
(Partial credit will be given for showing your work)

Problem 3 (4 points):

A composite torsion member consists of three concentric materials:

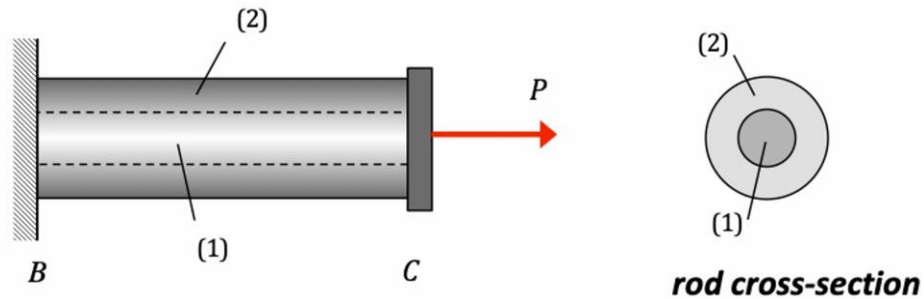
- ✧ Outer shell (1), with inner radius $1.5D$, outer radius $2D$, and shear modulus $G_1 = 2G$
- ✧ Intermediate shell (2), with inner radius D , outer radius $1.5D$, and shear modulus $G_2 = 3G$
- ✧ Solid core (3), with radius D and shear modulus $G_3 = 4G$

They are perfectly bonded (i.e., no slip, no relative sliding). When a torsional load T is applied. Which material experience the largest shear stress, (1), (2), or (3)?



Problem 4 (6 points):

A rod with length L is made of a shell (2) and core (1), where the areas are known to be $A_2 = 2A_1$ and the Young's modulus are known to be $E_1 = 3E_2$. The rod is fixed on the wall at end B, and the axial load P is acting on the rigid connector at end C. Let F_1 and F_2 denote the axial loads carried by the core (1) and shell (2), respectively.



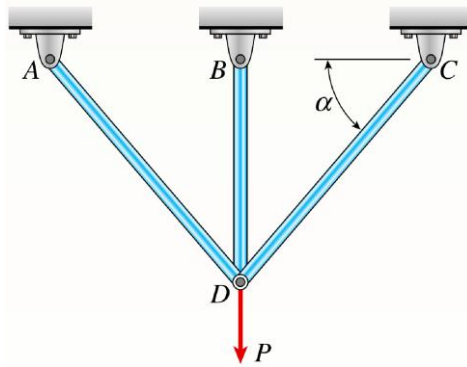
- (1) Determine whether the structure is statically determinate or indeterminate. (1 pt)
- If it is statically indeterminate, state the degree of static indeterminacy and write the compatibility equation(s). (2 pts)
 - If it is statically determinate, write the equilibrium equation(s). (2 pts)
- (3) Without solving for the exact values, determine which force has the larger magnitude, $|F_1|$ or $|F_2|$. (3 pts)

Problem 5 (6 points):

A symmetrical framework consists of three pin-connected cylindrical bars, as shown in the figure. The structure is subjected to a vertical load $P = 137.3 \text{ kN}$ applied at joint D. The two inclined bars, AD and DC, are made of steel with Young's modulus $E_s = 210 \text{ GPa}$. The vertical bar BD is made of aluminum with Young's modulus $E_a = 70 \text{ GPa}$. The angle between each inclined bar and the horizontal is $\alpha = 45^\circ$. The length of bar BD is $L = 1 \text{ m}$. All bars have the same circular cross section with radius $r = 1.0 \text{ cm}$.

- (a) What is the degree of static indeterminacy? (1 pt)
- (b) Write down the compatibility equation. (2 pts)
- (c) Determine the magnitude of the vertical displacement of point D, δ . (3 pts)

(Hint: $\sqrt{L^2 + (L + \delta)^2} \approx \sqrt{2}L + \frac{\delta}{\sqrt{2}}$ when δ is small)



Problem 6 (6 points):

A solid circular bar of diameter $d_1 = 25 \text{ mm}$ is enclosed by a tube of inner diameter $d_2 = 30 \text{ mm}$ and outer diameter $d_3 = 40 \text{ mm}$. Both bar and tube are held rigidly by a support at end A and joined securely to a rigid plate at end B. The composite bar, which has a length L , is twisted by a torque $T = 400 \text{ N} \cdot \text{m}$ acting on the end plate. The bar and tube have shear modulus of $G_1 = 200 \text{ GPa}$ and $G_2 = 100 \text{ GPa}$, respectively.

- (1) Please draw the FBD of the structure. (1 pt)
- (2) Determine the compatibility equation(s). (1 pt)
- (3) Calculate the polar moment of inertia of the tube and the inner bar. (2 pts)
- (4) Calculate the maximum shear stress τ_1 and τ_2 in the bar and tube, respectively. (2 pts)

